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STEPCONE-2025

STANDARD OPERATING PROCEDURE

EVENT NAME: Robo Race
DEPARTMENT NAME: ECE
EVENT TYPE: Flagship Event

EVENT DESCRIPTION:

Event consists of 2 rounds.

Round-1: This is the fundamental round for testing bots based on specified constraints, assessing their accuracy.

Round-2: It will be conducted in the race track, which features multiple levels of traps. The bot must navigate through all the traps and reach the destination in the shortest possible time.

ROUND-1:

Bots will undergo preliminary testing based on predefined constraints. Accuracy and control will be assessed. Teams will compete in a race tournament to qualify for the next round.

DESIGN SPECIFICATIONS:

Bot Dimensions & Weight :

- The Maximum dimensions of a bot: length 25 cm × breadth 25 cm × height 15 cm.
- Maximum weight: 2.5 kg (excluding the remote controller).
- All batteries must be onboard.

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MOBILITY:

Methods of mobility include:

- Rolling (wheels, tracks or the whole robot).

Mobility methods that are NOT allowed:

- Flying (using airfoil, helium balloons, etc.) is not allowed.

ROBOT CONTROL REQUIREMENTS:

- The robot must be controlled only through a **wireless** remote (Bluetooth or Wi-Fi module, etc.), while all power supply must be on board.
- External power sources (wired) are not permitted.
- The remote control system must be reliable and interference-free to ensure smooth operation.
- The team must pair the wireless remote with the bot before placing it in the arena. Once the bots are inside the arena, no additional time will be given for this process.

BATTERY AND POWER:

- The bot must be powered electrically. Onboard batteries must be sealed, immobilized-electrolyte types (such as lithium, NiCad, NiMH, or dry cells).
- The electric voltage of the bot should not exceed 12V DC .
- Participants will have to bring their own converters for standard power supply.
- Participants must protect the battery terminals from a direct short and causing a battery fire, failure to do so will cause direct disqualification.
- Use of damaged, non-leak-proof batteries may lead to disqualification.
- Special care should be taken to protect the onboard batteries. If the judges find that the battery is insufficiently protected, the team will be disqualified immediately.
- Change of battery will not be allowed during the match. If any bot stop in middle of the race 3min time is given to change the batteries by reducing 1 point.

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ROUND-2:

The bots are allowed to move inside the track that are designed which includes different levels. Creating a Roborace track using natural materials like mud, stones, and wood is a fun and eco-friendly way to challenge participants. Creating obstacles, boundaries, or checkpoints to make the path more interesting. Wooden planks or logs can be used as ramps, small bridges, or barriers to add height and tricky sections that test the robots control. The bot which passes the track in less time by following rules are the winners.

SAFETY RULES:

- **Participant Responsibility:** All participants must ensure their bots with design specifications and safety regulations. Any violation will lead to disqualification.
- **Supervision:** The event will be monitored by judges and event coordinators to ensure fair play and safety compliance.
- **No External Assistance:** Teams are not allowed to seek external help during the event. Only registered team members may operate and modify the bot.
- **Arena Conduct:** Participants must not enter the race track unless instructed by the event organizers.
- **Prohibited Substances:** Flammable, explosive, or hazardous materials are strictly prohibited in the bots.
- **Respect for Equipment:** Any damage to event property (arena, track, or equipment) due to negligence will lead to disqualification.
- **Emergency Handling:** In case of an emergency (fire, battery failure, or circuit shorting), participants must immediately inform the organizers.

EVENT GUIDELINES:

- A team should consist of 2-4 participants.
- **Team Representative:** Each team must specify their team representative (leader) at the time of registration on the website. All the important communications between Organizers and the registered teams will be done through their team representative. The team representative must submit valid contact details (phone no., email ID etc.) at the time of registration.

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- Team is disqualified if the bot is not built with the mentioned constraints.
- Participants must be students currently pursuing Undergraduate (UG) or Postgraduate (PG) studies.

Team Size/No of Participants per team: 2-4 members

PLAN OF ACTION: - (day-shift)

Registration fee- Rs.1000/- per team

Judging Criteria:

- Teams filtered based on performance and time criteria will be qualified to second round .
- The team that reaches the destination in the shortest time will be declared the winner, while the other team will be the runner-up.
- All participants will receive a certificate of participation.